

Object Follow

Package: `microros_v2_kcf_follow_pkg`

1. Feature Description

`microros_v2_kcf_follow_pkg` is a follow controller based on the target bounding box. It only subscribes to the target box published by a tracking node, and then controls the pan-tilt and/or the body motion.

Therefore, the complete functionality consists of two parts:

1. A tracker obtains the target bounding box from the image, e.g. `smart_tracker/super_tracker` or the KCF tracking node.
2. `tracking_bbox_follow_node` computes the pan-tilt angle, the body angular velocity, and the body linear velocity from the target box center and the target box size.

The default control rate is 10 Hz. When the target box is invalid, its confidence is too low, or it has not been updated for longer than `target_stale_timeout_sec`, the node immediately publishes zero velocity; the source code has no "allow 5 more frames of movement" lost-target tolerance logic.

2. Control Chain

```
Camera / vision tracker
├── /tracking_bbox
│   └── tracking_bbox_follow_node (10 Hz)
│       ├── PTZ follow: target center -> /cmd_ptz_angle
│       ├── Body turning: target horizontal error or pan-tilt angle ->
│       │   /cmd_vel.angular.z
│       └── Distance follow: target box size error -> /cmd_vel.linear.x
```

PTZ Version

`enable_ptz_follow:=true` by default:

- Based on the pixel error of the target center relative to the image center, independent X/Y PD control updates the pan-tilt angles.
- Once the pan-tilt horizontal angle deviates from the initial angle by `servo1_engage_deg`, the body PD control drives `angular.z`, letting the body gradually follow the pan-tilt back to center.
- `linear.x` can be controlled simultaneously based on the target box size.

No-PTZ Version

A fixed camera cannot adjust its line of sight through the pan-tilt, so at startup you must set:

```
enable_ptz_follow:=false publish_initial_ptz_command:=false
```

In this mode the node uses the target center's horizontal pixel error to control the body `angular.z`, and controls `linear.x` based on the target box size. Setting only `publish_initial_ptz_command:=false` does not disable the PTZ follow logic, so it cannot be the only change for a No-PTZ configuration.

3. Topics and Messages

Topic	Direction	Message Type	QoS	Description
<code>/tracking_bbox</code>	Subscribe	Auto-discovered by default	<code>BEST_EFFORT</code> , depth=1	Target bounding box input
<code>/cmd_vel</code>	Publish	<code>geometry_msgs/msg/Twist</code>	<code>BEST_EFFORT</code> , depth=1	<code>linear.x</code> forward/back, <code>angular.z</code> left/right turn
<code>/cmd_ptz_angle</code>	Publish	<code>geometry_msgs/msg/Vector3</code>	depth=10	Published in PTZ mode; <code>x</code> is the horizontal angle, <code>y</code> is the tilt angle, in degrees
<code>/JoyState</code>	Subscribe	<code>std_msgs/msg/Bool</code>	depth=10	When <code>true</code> , pauses auto-follow and publishes zero velocity

The node is also compatible with:

- `smart_tracker/msg/TrackBox`, auto-discovered via `tracking_bbox_msg_type:=auto`;
- `std_msgs/msg/Int32MultiArray`: when the length is at least 11, it is parsed as `[valid, x1, y1, x2, y2, cx, cy, width, height, image_width, image_height]`; when the length is 4 to 10, the first four elements are taken as the bounding box;
- messages with `x1`, `y1`, `x2`, `y2` fields.

The bounding box format is uniformly the top-left `(x1, y1)` and bottom-right `(x2, y2)`. The node auto-sorts the coordinates and checks the width, height, and confidence.

4. Startup Steps

The following commands assume the workspace is `~/microros_car_v2_ws` and the ROS 2 environment is already configured.

4.1 Start the Chassis, Agent, Camera, and Model

Terminal 1:

```
ros2 launch microros_v2_bringup car_base.launch.py
```

```

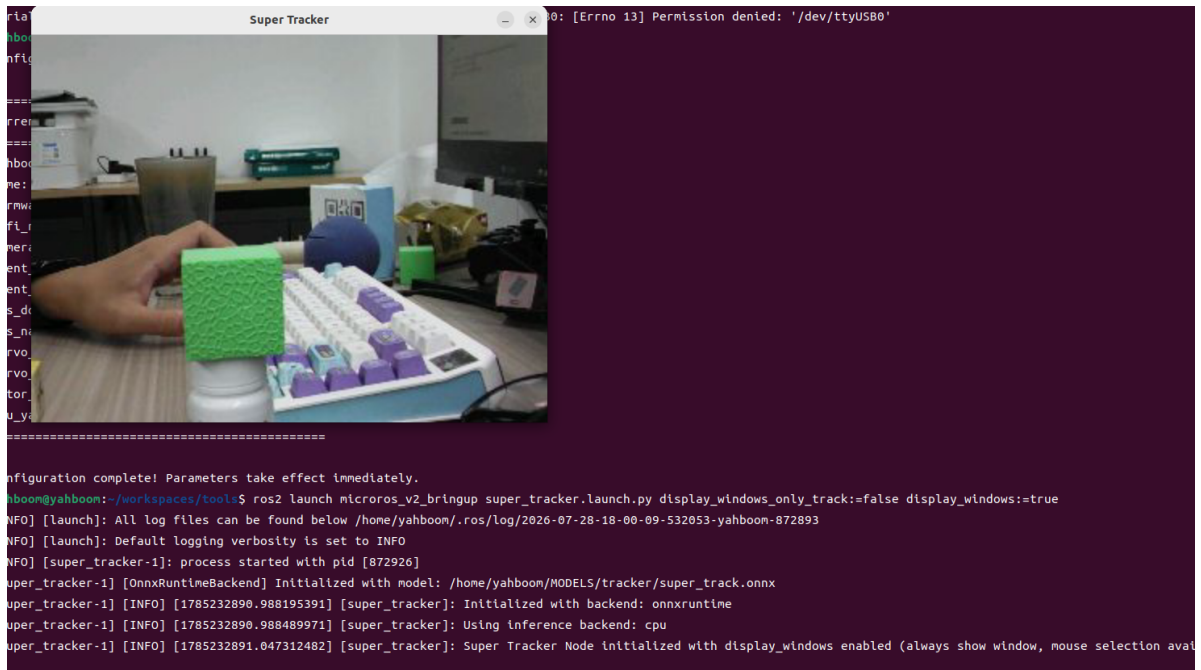
yahboom@yahboom:~$ ros2 launch microros_v2_bringup car_base.launch.py
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-07-28-15-04-46-269642-yahboom-579370
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [micro_ros_agent-1]: process started with pid [579411]
[INFO] [camera_driver-2]: process started with pid [579413]
[INFO] [robot_state_publisher-3]: process started with pid [579415]
[INFO] [joint_state_publisher-4]: process started with pid [579417]
[INFO] [ekf_node-5]: process started with pid [579419]
[micro_ros_agent-1] [1785222287.404236] info | UDPv4AgentLinux.cpp | init | running... | port: 8090
[robot_state_publisher-3] [INFO] [1785222288.847816851] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-3] [INFO] [1785222288.848013042] [robot_state_publisher]: got segment base_link
[robot_state_publisher-3] [INFO] [1785222288.848028942] [robot_state_publisher]: got segment bwheel_Link
[robot_state_publisher-3] [INFO] [1785222288.848037122] [robot_state_publisher]: got segment cam1_Link
[robot_state_publisher-3] [INFO] [1785222288.848043874] [robot_state_publisher]: got segment cam2_Link
[robot_state_publisher-3] [INFO] [1785222288.848050530] [robot_state_publisher]: got segment cam3_Link
[robot_state_publisher-3] [INFO] [1785222288.848057264] [robot_state_publisher]: got segment imu_frame
[robot_state_publisher-3] [INFO] [1785222288.848063747] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-3] [INFO] [1785222288.848070240] [robot_state_publisher]: got segment lfwheel_Link
[robot_state_publisher-3] [INFO] [1785222288.848076888] [robot_state_publisher]: got segment rfwheel_Link
[joint_state_publisher-4] [INFO] [1785222289.372349515] [joint_state_publisher]: Waiting for robot_description to be published on the robot_description
[camera_driver-2] [INFO] [1785222289.716161193] [esp32_ip_camera_publisher]: camera node started, camera_url: http://192.168.12.37:81/stream
[camera_driver-2] [WARN] [1785222292.785741414] [esp32_ip_camera_publisher]: Camera not opened, trying reconnect... (next retry in 1.0s)
[camera_driver-2] [INFO] [1785222294.072701944] [esp32_ip_camera_publisher]: IP camera stream opened successfully

```

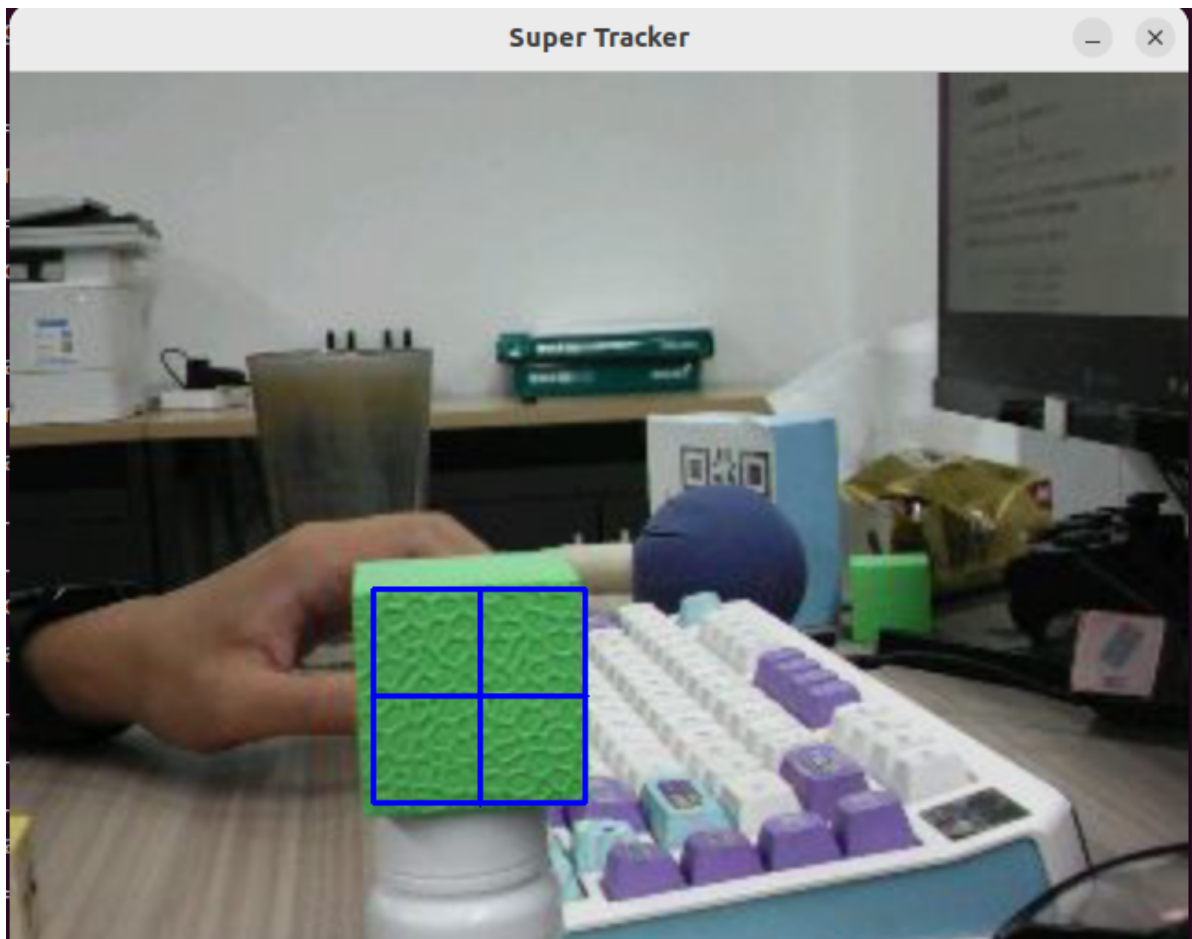
4.2 Start the Target Tracker

Run `smart_tracker/super_tracker`, Terminal 2:

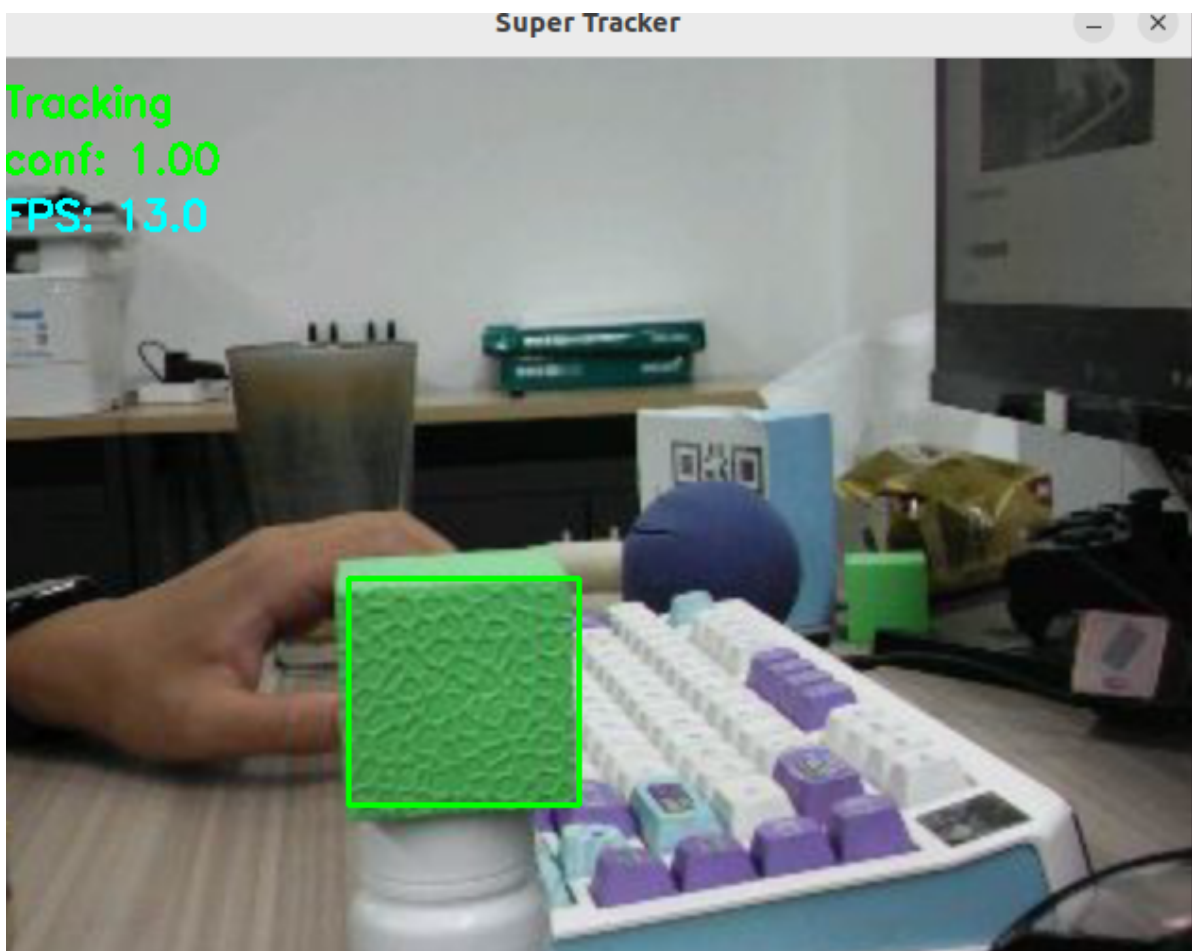
```
ros2 launch microros_v2_bringup super_tracker.launch.py
```



Press `C` to enter the box-selection mode, then select with the mouse:



Then press Enter to start following the coordinate data:



4.3 Start the Follow Controller

Terminal 3:

```
ros2 launch microros_v2_kcf_follow_pkg tracking_bbox_follow.launch.py
```

After a successful startup, slowly move the object for left/right tracking and forward/back following:

```
yahboom@yahboom: $ ros2 launch microros_v2_kcf_follow_pkg tracking_bbox_follow.launch.py \
enable_ptz_follow:=false \
publish_initial_ptz_command:=false
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-07-28-18-04-42-161871-yahboom-877278
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [tracking_bbox_follow_node-1]: process started with pid [877299]
[tracking_bbox_follow_node-1] [INFO] [1785233086.913368656] [tracking_bbox_follow_node]: Subscribed /tracking_bbox, type=smart_tracker/msg/TrackBox
[tracking_bbox_follow_node-1] [INFO] [1785233086.913783152] [tracking_bbox_follow_node]: Tracking bbox follower started. bbox_topic=/tracking_bbox, cmd_vel_topic=/cmd_vel
ptz=False, body_turn=True, distance_follow=True
[tracking_bbox_follow_node-1] [WARN] [1785233086.933133416] [tracking_bbox_follow_node]: No valid bbox; stopping. valid=False fresh=False confidence=0.000
[tracking_bbox_follow_node-1] [INFO] [1785233087.091385553] [tracking_bbox_follow_node]: Target bbox size initialized from first valid bbox: 123.5 px
```

5. Distance-follow Algorithm

The node does not use the color radius to estimate distance; instead it uses the geometric size of the bounding box:

```
current_size = sqrt(width * height)
size_error = (target_size - current_size) / target_size
```

`auto_set_target_bbox_size:=true` by default. The first valid bounding box received after startup is used as the target distance baseline. You can also specify `target_bbox_size_px` manually, in which case the node no longer waits for the first bounding box to initialize the baseline.

Distance control uses the PID form, with `distance_ki:=0.0` by default, so it is actually PD:

```
linear.x = linear_dir * (distance_kp * size_error
                        + distance_ki * integral
                        + distance_kd * d_error)
```

- When the target box is smaller than the baseline, `size_error > 0`, the default is to move forward and approach;
- When the target box is larger than the baseline, `size_error < 0`, the default is to move backward and move away;
- When `abs(size_error) <= distance_tolerance_ratio`, forward/back movement stops;
- `linear.x` is limited to `[-max_linear_x, max_linear_x]`, with the default maximum absolute value being `0.12 m/s`.

6. Key Parameters

Common Parameters

Parameter	Default Value	Description
<code>tracking_bbox_topic</code>	<code>/tracking_bbox</code>	Target box input topic; changing it requires a node restart
<code>tracking_bbox_msg_type</code>	<code>auto</code>	Auto-discovers the message type; a full ROS type name can also be filled in
<code>cmd_vel_topic</code>	<code>/cmd_vel</code>	Body velocity output topic; changing it requires a node restart
<code>ptz_command_topic</code>	<code>/cmd_ptz_angle</code>	PTZ angle output topic; changing it requires a node restart
<code>joy_state_topic</code>	<code>/JoyState</code>	Gamepad takeover state topic; changing it requires a node restart
<code>enable_follow</code>	<code>true</code>	Master follow switch; when off, zero velocity is published continuously
<code>enable_ptz_follow</code>	<code>true</code>	Enable pan-tilt following; No-PTZ must be set to <code>false</code>
<code>enable_body_turn_follow</code>	<code>true</code>	Enable body horizontal turning
<code>enable_distance_follow</code>	<code>true</code>	Enable forward/back distance following
<code>enable_joy_override</code>	<code>true</code>	Pause auto-follow when the gamepad takes over
<code>image_width</code> / <code>image_height</code>	<code>640</code> / <code>480</code>	Used when the input message does not provide the image size
<code>tracking_confidence_threshold</code>	<code>0.1</code>	Minimum valid confidence
<code>target_stale_timeout_sec</code>	<code>1.2</code>	Stop when the target box has not been updated for longer than this time
<code>follow_rate_hz</code>	<code>10.0</code>	Control timer rate, minimum 1 Hz

Pan-Tilt and Body Turning Parameters

Parameter	Default Value	Description
<code>pixel_tolerance</code>	15.0 px	PTZ horizontal/tilt pixel deadband
<code>kp_x</code> / <code>kd_x</code>	0.6 / 0.15	PTZ horizontal PD coefficients
<code>kp_y</code> / <code>kd_y</code>	0.6 / 0.15	PTZ tilt PD coefficients
<code>max_step_xy</code>	5.0 deg	Maximum pan-tilt angle step per control cycle
<code>horizontal_min_deg</code> / <code>horizontal_max_deg</code>	-90 / 90 deg	Horizontal angle limits
<code>pitch_min_deg</code> / <code>pitch_max_deg</code>	-90 / 20 deg	Tilt angle limits
<code>initial_horizontal_deg</code>	0.0 deg	Initial horizontal angle, also the body homing reference angle
<code>initial_pitch_deg</code>	-45.0 deg	Initial tilt angle
<code>body_kp</code> / <code>body_kd</code>	0.03 / 0.0	PTZ-mode body turning PD coefficients
<code>servo1_engage_deg</code>	5.0 deg	Body turning starts when the pan-tilt horizontal deviation reaches this value
<code>body_tolerance_deg</code>	3.0 deg	Body homing stop deadband
<code>max_angular_z</code>	0.6 rad/s	Maximum absolute body angular velocity
<code>body_ff_gain</code>	1.0	In PTZ mode, the feedforward coefficient compensating the pan-tilt horizontal angle from the body angular velocity
<code>fixed_camera_pixel_tolerance</code>	15.0 px	No-PTZ body turning stop deadband
<code>fixed_camera_engage_px</code>	30.0 px	No-PTZ body turning start threshold
<code>fixed_camera_body_kp</code> / <code>fixed_camera_body_kd</code>	0.6 / 0.15	No-PTZ body turning PD coefficients
<code>body_dir</code> / <code>fixed_camera_body_dir</code>	-1.0 / -1.0	Body turning direction correction; change to 1.0 if necessary

Parameter	Default Value	Description
<code>sign_x</code> / <code>sign_y</code>	1.0 / -1.0	PTZ horizontal/tilt direction correction

Distance-follow Parameters

Parameter	Default Value	Description
<code>target_bbox_size_px</code>	0.0	Target baseline size; when 0, set automatically from the first valid box
<code>auto_set_target_bbox_size</code>	true	Whether to initialize the baseline size from the first valid box
<code>distance_tolerance_ratio</code>	0.06	Relative size deadband, 6% by default
<code>distance_kp</code> / <code>distance_ki</code> / <code>distance_kd</code>	0.18 / 0.0 / 0.02	Distance PID coefficients
<code>distance_integral_limit</code>	0.5	Integral limit
<code>max_linear_x</code>	0.12 m/s	Maximum absolute linear velocity
<code>linear_dir</code>	1.0	Forward/back direction correction; change to -1.0 if necessary